

Element F

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How does the design solve the problem of the usable practice device for a college class?:

1. Portable:

- a. Since the device is just a cylinder that can be submerged in water, it can be easily used by many people and transported around.
- b. Credible evidence: Each unit of this device can be taken apart, moved around, and safely stored with no issues.

2. Reusable:

- a. We bought many cheap mouthpiece filters. They can easily be removed and replaced with a clean one. The ease of switching out the mouthpiece and its mobility allow many people to try it quickly.



- b. Credible evidence:

3. Safe

- a. There are no dangerous aspects to this design. The only possible danger to be careful of is the inhaling of water. If the water gets too close to the mouthpiece, it would be wise to remove your mouth from the mouthpiece.

- b. Credible evidence: We purposely ensured that in our design, there will be no sharp edges on this device. We also made sure to carefully calculate the pipe's height and take into consideration how much you can suck in without inhaling the water.

4. User- Friendly

- a. The device is easy to use: Just turn the base to the desired setting and submerge it in water. The students should know how to handle it, as the experts will not always be available to help.
- b. Credible evidence: As a group we have all tested this design multiple times and had no issues with how the device works and being confused on what to do

5. Precise

- a. Precision is necessary for a reliable device to measure and store reliable data.
- b. Credible evidence: We have used the device a few times, and the only variability is in slight fluctuations in atmospheric pressure

6. Inexpensive

- a. Dr. Brossman has a classroom of 90+ students. Buying multiple devices from an inexpensive source will save Dr. Brossman and the University a lot of money.

b.

